LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Practical demonstration of steady streamlined flow pattern over object in a smoke tunnel.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Practical demonstration of steady streamlined flow pattern over object in a smoke tunnel.
5. Specific Objectives	11.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of smoke tunnel flow patterns over NACA airfoil; Safety: Ensure smoke density is low to prevent inhalation; Supervision: Instructor monitors flow stability and trainee observation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three primary properties of the atmosphere affecting flight?
(b) Define the term 'boundary layer' in the context of aerodynamics.
(c) How does fluid velocity change as it passes over a streamlined body?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to Smoke Tunnel Mechanics • Generation of Steady Streamlined Flow • Visualization Techniques using Smoke • Flow Separation and Reattachment Points • Impact of Object Shape on Flow Pattern	L & Ds	PPT, IB	15 min
(b)	• Analysis of Boundary Layer Behavior • Identification of Separation Zones • Effect of Angle of Attack on Flow • Comparison of Streamlined vs Non-Streamlined Objects • Recording Observations of Flow Characteristics	L & Ds	PPT, PDF	20 min
(c)	• Interpretation of Flow Visualization Data • Correlation with Theoretical Flow Patterns • Discussion on Practical Applications • Review of Safety Protocols Used • Preparation for Next Lesson Video Analysis	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the boundary layer behaves over a streamlined body in steady flow.
(b) Describe the visual indicators of flow separation observed in the smoke tunnel demonstration.
(c) How does the shape of the object influence the stability of the flow pattern?

12. Home Assignment and Follow Up	Prepare a brief note on how the flow patterns observed in this lesson relate to the video demonstration of fluid flow characteristics in the next lesson plan.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Video demonstrating fluid flow characteristics. Video on theory of flight	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min